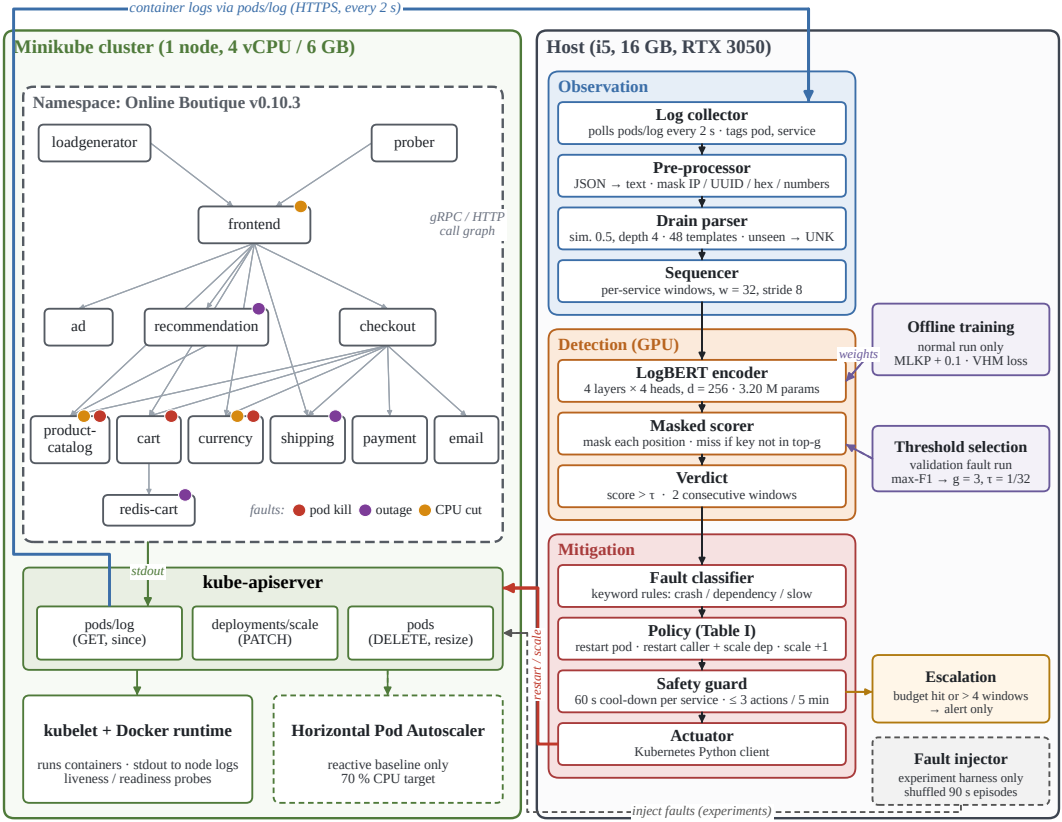

container logs via pods/log (HTTPS, every 2 s)
Minikube cluster (1 node, 4 vCPU / 6 GB)
Namespace: Online Boutique v0.10.3
loadgenerator
prober
frontend
gRPC / HTTP call graph
ad
recommendation
checkout
product-catalog
cart
currency
shipping
payment
email
redis-cart
faults: pod kill outage CPU cut
stdout
kube-apiserver
pods/log (GET, since)
deployments/scale (PATCH)
pods (DELETE, resize)
kubelet + Docker runtime
runs containers · stdout to node logs
liveness / readiness probes
Horizontal Pod Autoscaler
reactive baseline only
70 % CPU target
Host (i5, 16 GB, RTX 3050)
Observation
Log collector
polls pods/log every 2 s · tags pod, service
Pre-processor
JSON → text · mask IP / UUID / hex / numbers
Drain parser
sim. 0.5, depth 4 · 48 templates · unseen → UNK
Sequencer
per-service windows, w = 32, stride 8
Detection (GPU)
weights
LogBERT encoder
4 layers × 4 heads, d = 256 · 3.20 M params
Masked scorer
mask each position · miss if key not in top-g
Verdict
score > τ · 2 consecutive windows
Offline training
normal run only
MLKP + 0.1 · VHM loss
Threshold selection
validation fault run
max-F1 → g = 3, τ = 1/32
Mitigation
Fault classifier
keyword rules: crash / dependency / slow
Policy (Table I)
restart pod · restart caller + scale dep · scale +1
Safety guard
60 s cool-down per service · ≤ 3 actions / 5 min
Actuator
Kubernetes Python client
Escalation
budget hit or > 4 windows
→ alert only
Fault injector
experiment harness only
shuffled 90 s episodes
restart / scale
inject faults (experiments)